

Concepts & Corrections

A consolidated, visual summary of the methodological corrections in Experiment A and the core concepts behind the whole project — with the numbers and the mathematics that make each claim checkable

1. The corrections: what changed between run 1 and run 2, and why

Parameter	Run 1	Run 2 (final)	Reason
Samples	1,646	10,000	1.6 samples per input dimension starves the fit
WikiText split	validation	train	the validation split simply ran out: only ~1,646 passages over 100 chars exist in it
Everything else	—	unchanged	same models, control, pooling, ridge penalty, 75/25 split, metrics, thresholds

1.1 The governing arithmetic

Stage A3 fits, per layer pair, a 768x768 map. Its closed form is:

$$W = (X^T X + aI)^{-1} X^T Y$$

The inverted matrix depends on the inputs alone — the outputs never enter it — so the fit **decomposes into 768 independent regressions**, one per output dimension, sharing one design matrix and one inverse. The statistical requirement is therefore rows per **input dimension**, not rows per total parameter count:

Run	Training rows	Rows / 768 (governs)	Rows / 590K params (misleading)
1	1,234	1.6 — starved	0.002
2	7,500	9.8 — adequate (rule of thumb: 5–10)	0.013

Consequence: output width is nearly free — the 512-to-768 adapter of Experiment B needs rows relative to 512, which is why 3,000 image pairs sufficed there.

1.2 Why starvation biases downward, not randomly

Through the SVD lens, ridge multiplies each data direction by $s/(s^2+a)$ instead of OLS's $1/s$. With ample data, most directions have s large and pass through; with 1.6 rows per dimension most directions are weak, shrinkage dominates everywhere, and the fit systematically **under-predicts**. The tell in run 1: the random control scored $R^2 = -0.773$ on held-out rows — generalizing worse than predicting the mean, the signature of a memorized-then-shrunk fit, not of absent structure.

1.3 Train vs validation — why the switch carries no leakage

The split roles (train = weights fitted on it; validation = guides the experimenter's tuning; test = used once for the honest number) presuppose **a model being trained**. Experiment A trains nothing: GPT-2 and Pythia are frozen and the text is **probe stimuli** — inputs used only to elicit activations, like a stimulus set in a perception experiment. Both splits are the same Wikipedia corpus, partitioned at the article level and cleaned identically; they differ only in which articles landed in which file. The held-out discipline that actually matters lives one level down: A3 holds out 25% of the samples, and every reported R^2 is measured on rows the ridge never saw. One real asymmetry is disclosed for completeness: The Pile (Pythia's training data) includes Wikipedia while GPT-2's WebText excluded it — but this applies identically to both splits, the random control sees the same text, and the hypothesis concerns whether a linear map exists, not how surprising the input was.

1.4 One threshold recorded as missed

The pre-registered CKA criterion said 'diagonal mean > 0.5'; the final run reached 0.424, with a random baseline of 0.067 and a ratio of 6.3x. The absolute threshold was set optimistically — absolute CKA between models with different tokenizers is expected to be modest — and the ratio to the control is the

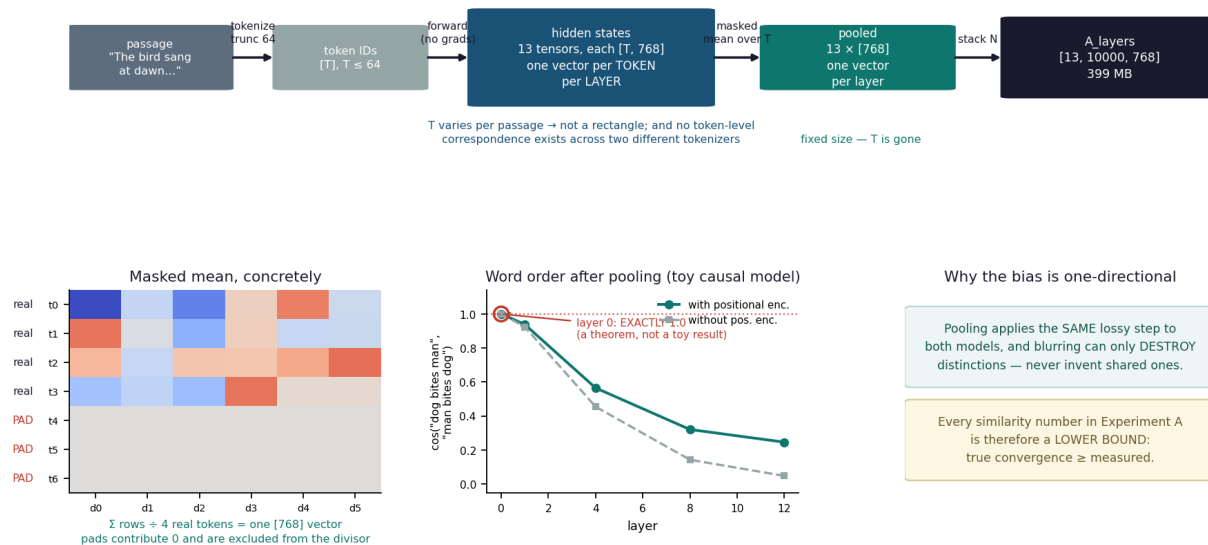
load-bearing test. Recorded as missed rather than retro-fitted; the verdict rests on the ratio and stitching criteria, which passed decisively.

2. The vocabulary, each term pinned to a measured number

Term	Meaning in one line	Where it is measured here
Embedding / representation	a vector positioning an input in a space where distance ~ dissimilarity	768-d LLM states; 512-d MobileCLIP; 768-d SigLIP
Activation / hidden state	a layer's output for an input; one per token per layer	A_layers [13, 10000, 768]
Mean pooling	masked average over tokens -> one vector per text; lossy	layer-0 order loss is exact (cos = 1.0000 for reversed sentences)
L2 normalization	scale to unit length; afterwards dot = cosine	31.92 raw dot vs 0.0355 cosine = dot-after-L2
Cosine similarity	angle-based similarity, blind to length	the metric Qdrant uses; adapter judged by it (0.900)
Anisotropy	uneven variance across directions	eff. dim 62.9 -> 32.9 under stretch; the reason Procrustes failed
Procrustes	best pure rotation between paired point sets (SVD: $R = UV^T$)	$R^2 = 0.038$ vs ridge 0.592 (Exp B)
Ridge regression	linear fit with an aI stabilizer; closed form, deterministic	every adapter and stitching map in the project
SVD	any linear map = rotate x scale x rotate	explains Procrustes (discards S) and the 1024->512 null space
Isomorphic up to linear map	same information, different coordinates, one matrix converts	the hypothesis; confirmed at R^2 0.74-0.80 (Exp A)
Permutation symmetry	reorder neurons + rewire: same function, unrelated weights	output diff 0.0, weight cosine -0.125 (demonstrated)
Samples per dimension	training rows / input width; the power measure for a linear fit	1.6 (run 1, failed) vs 9.8 (run 2, passed)
Probe stimuli	inputs fed to frozen models only to read activations	the 10,000 WikiText passages
Held-out evaluation	score on rows the fit never saw	all reported R^2 / recall; control at -0.773 caught memorization

3. From one passage to one row of the array

From one passage to one row of the saved array



Why pool at all — three reasons, one decisive

1 · ALIGNMENT (decisive)	2 · FIXED SHAPE	3 · SIZE
Different tokenizers → different token sequences of different lengths. Token-to-token correspondence does not exist; passage-to-passage does. Pooling moves the comparison to the level where alignment is real.	T varies per passage, so per-token states cannot form a rectangle. CKA and ridge both need [N, 768] matrices with row i = passage i in every model.	pooled: 3 × 399 MB = 1.20 GB per-token: 3 × 25.6 GB = 76.7 GB (T = 64 → 64× more) 1.2 GB fits Drive and loads in seconds.

Figure 1: The extraction flow, the masked mean, how much word order survives pooling, and why pooling at all.

Storage arithmetic: one pooled vector = $768 \times 4 \text{ B} = 3 \text{ KB}$; one model = $13 \times 10,000 \times 3 \text{ KB} = 399 \text{ MB}$; three models = 1.20 GB. Per-token storage would be 64x larger (76.7 GB at $T = 64$) — and useless anyway, since the two tokenizers admit no token-level correspondence. The layer-0 point in the middle panel is exact rather than simulated: an embedding lookup has no context, so pooling averages a multiset, and averaging is permutation-invariant — 'dog bites man' and 'man bites dog' pool to the identical vector, cosine 1.0000. Deeper layers separate them ($0.94 \rightarrow 0.25$ in the toy causal model) because causal attention makes each token's vector depend on what preceded it. Crucially the loss is one-directional: pooling can only blur distinctions, never invent shared ones, so every similarity in Experiment A is a lower bound on the true convergence.

4. The geometry, demonstrated with numbers

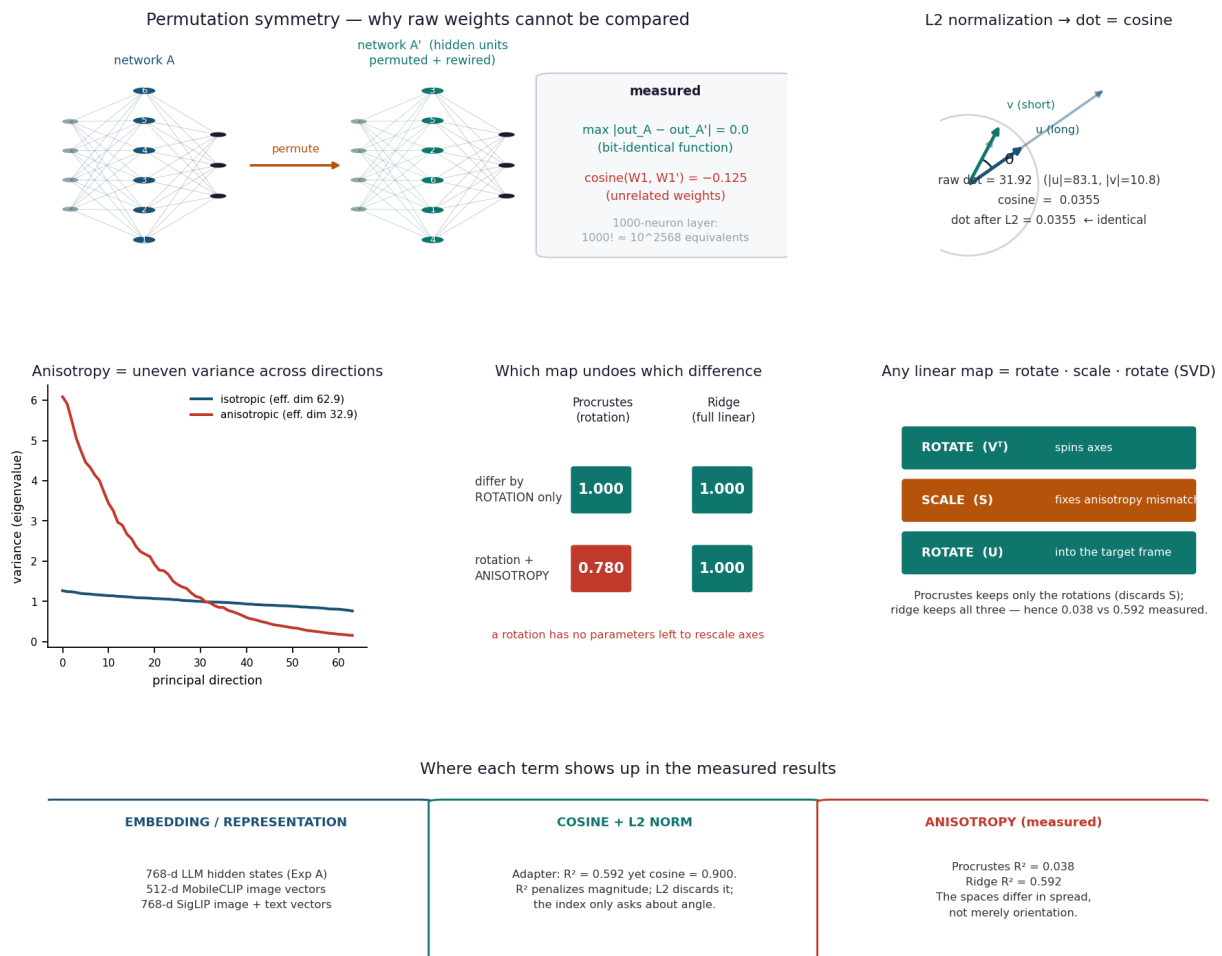


Figure 2: Permutation symmetry, L2/cosine, anisotropy, Procrustes-vs-ridge, and the SVD decomposition — every number computed, not asserted.

The Procrustes-vs-ridge panel is the mechanism behind the project's measured 0.038-vs-0.592: when two spaces differ by rotation only, both maps score 1.000; add anisotropy and Procrustes drops (0.780 in the synthetic case) while ridge stays at 1.000, because an orthogonal matrix has all singular values equal to 1 and no capacity left to rescale axes. The SVD panel shows why ridge succeeds: any linear map factors into rotate x scale x rotate, and ridge keeps all three factors where Procrustes keeps only the rotations. One honest confound from Experiment B carries over: the 512-vs-768 dimension gap forced zero-padding, which mechanically handicaps rotation as well — the same-dimensional follow-up in the report's Section 5 is designed to separate the two causes.

5. What the adapter does to the index geometry

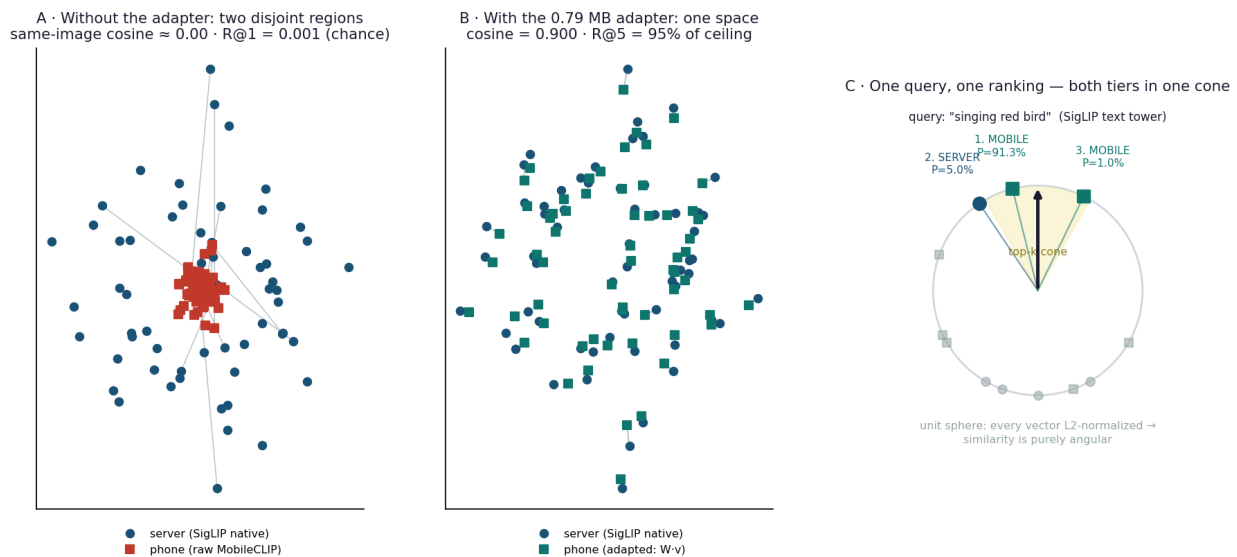


Figure 3: Before the adapter, with it, and the query cone. Points are illustrative but constructed to reproduce the measured statistics (adapted cosine 0.900; raw cross-space cosine ~ 0.00).

Panel A is the disease: same-image pairs sit far apart, so cross-tier search scores exact chance ($R@1 = 0.001$). Panel B is the cure: one 0.79 MB matrix, and each phone vector lands on its server twin — the clouds interleave, with no region one could circle and call 'the mobile part'. Panel C is retrieval: after L2 normalization every vector lives on the unit sphere, a query is a direction, and top-k is the narrowest cone containing k vectors — the measured demo ranking (MOBILE 91.3% / SERVER 5.0% / MOBILE 1.0%) sits inside one cone. This panel also dissolves an apparent contradiction: the adapter's R^2 of 0.592 and cosine of 0.900 coexist because R^2 penalizes magnitude errors that normalization discards, while cosine measures the only thing the index ever asks — direction.

6. Layer correspondence, and where the evidence actually lives

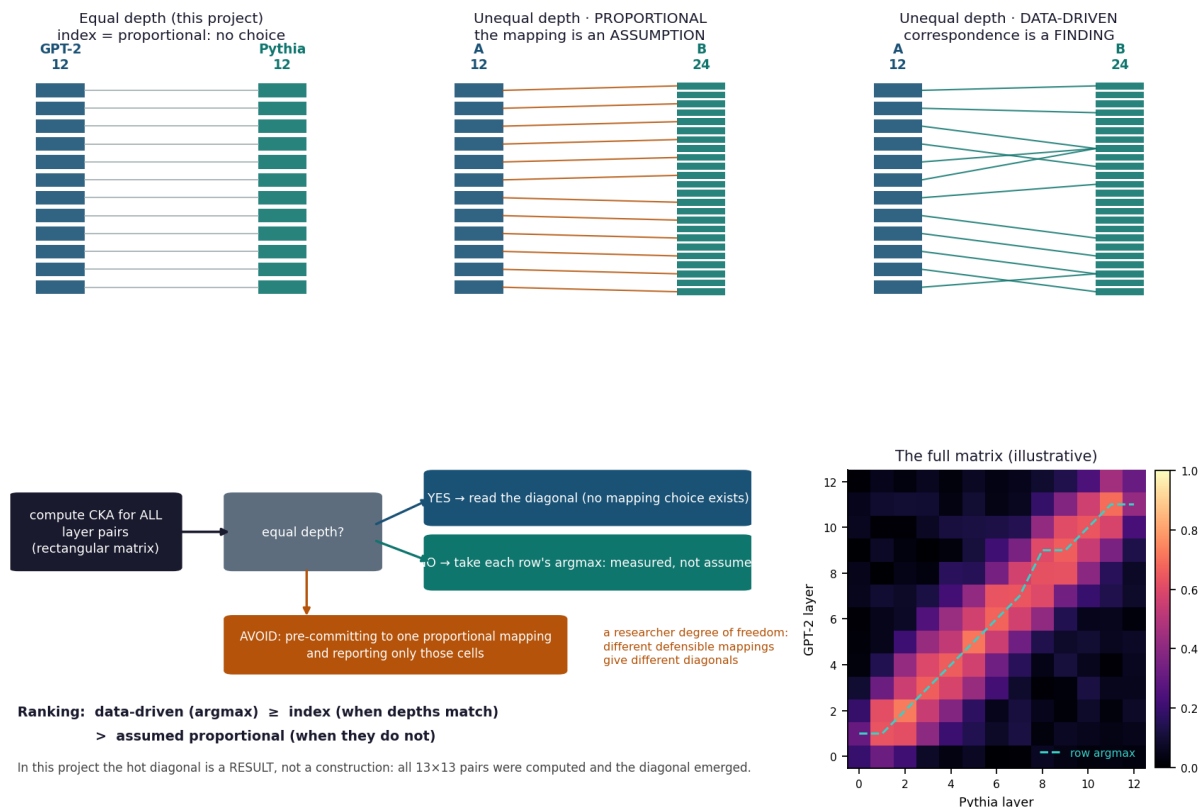


Figure 4: Three correspondence strategies, the decision flow, and the full matrix with its row-argmax overlay.

Both models happen to have 12 blocks (the BERT-base convention: $\sim 12d^2$ parameters per block, so 12 blocks at $d = 768$ is $\sim 85M$ — plus embeddings, the familiar 124M), which makes index matching and proportional matching coincide, so no mapping choice had to be defended. Had the depths differed, the right move is not an assumed proportional mapping but the data-driven one: compute all pairs and let each row's argmax say which layers correspond.

The deeper point generalizes. 'We matched by index and the numbers were high' reports 13 measurements and is silent about the other 156 cells — every cell might be high (no structure), or the true best matches might sit off-diagonal (offset correspondence). 'We computed all 169 pairs and the diagonal lit up' is a claim that could have failed in visible ways — a uniformly bright square, a shifted band, a blocky pattern — and did not. The information is in the **contrast**, not the diagonal's height. The same structure carries the whole project: R^2 0.737 means something against a control at 0.173; the MLP's 0.903 means something against the linear map's 0.900; a rank-1 demo retrieval means something beside a negative control that stays flat. In every case, the second measurement is the one that could have come out otherwise — which is exactly where the evidence lives.

7. Experimental design vocabulary (from the cheat sheet)

Term	Meaning
Train / validation / test split	Partitions defined by how much information reaches the model: train (weights fitted on it), validation (guides the experimenter's choices), test (used once, for the honest final number). The roles presuppose a model being trained - they carry no meaning for frozen models used only to read activations.
Probe stimuli	Inputs fed to a frozen model purely to elicit internal activations for measurement, analogous to a stimulus set in a perception experiment. WikiText plays this role in Experiment A: it trains nothing.
Held-out evaluation	Measuring a fitted map on rows it never saw. A 768x768 matrix with 590K free parameters can memorize a small sample exactly; in-sample R^2 would then be meaningless. Every R^2 in this project is held-out.
Samples per dimension	Training rows divided by input dimensionality - the practical power measure for a linear fit. Below roughly 5, ridge shrinkage dominates and estimates become systematically pessimistic. Run 1: 1.6. Run 2: 9.8.
Shrinkage bias (starved regression)	When rows are scarce relative to parameters, the ridge penalty forces coefficients toward zero to stay stable. The fit then understates the true relationship - one-directionally. Negative held-out R^2 on a control is its signature.
Pre-registration	Fixing success criteria before seeing results, so a verdict cannot be chosen after the fact. Used for both experiments; Section 1.4 records the one sub-threshold that was missed.
Diagnosed confound	A specific, checkable reason a measurement may be wrong, identified before rerunning and paired with a prediction that distinguishes it from the alternative. What separates a correction from fitting the result.

Rule of thumb. Before accepting any negative result from a fitted map, check that the measurement had the power to show a positive one: rows per parameter first, then alignment of the two data matrices, then the pooling or preprocessing that produced them.

8. Anisotropy, measured in this project's own activations

The geometry section above demonstrates anisotropy on synthetic data. A direction-symmetry check on Experiment A produced the same phenomenon in the real activations, and quantified it. Effective dimensionality below is the participation ratio of the covariance spectrum - the number of directions that effectively carry the variance, out of 768 available:

Layer	GPT-2 effective dim	Pythia effective dim	Ratio
0	8.9	92.7	10.4x
4	3.3	9.3	2.8x
8	10.9	19.8	1.8x
12	2.2	43.9	20.0x

GPT-2's variance concentrates into as few as two or three effective directions - the documented rogue-dimension phenomenon, measured here directly. Two consequences followed, both verified:

It inflates variance-weighted scores when GPT-2 is the target. Fitting the reverse map (Pythia -> GPT-2) appeared to beat the forward map in middle layers by 0.08-0.12 R^2 . Recomputed with uniform averaging across output dimensions, the mean absolute gap fell from 0.097 to 0.006 and the directions became indistinguishable (layer 4: 0.722 forward, 0.723 reverse). The apparent asymmetry was the metric, not the models.

It is the same property that defeats rotation in Experiment B. A rotation preserves variance spread by construction, so it cannot reconcile two spaces whose spread differs - which is why Procrustes scored 0.038 where ridge scored 0.592. One measured geometric property therefore explains an artifact in the scientific experiment and a design decision in the engineering one.

A genuine information difference, separately identified. At layer 0 the asymmetry runs the other way and is not an artifact: sequence length is recoverable from GPT-2's pooled layer-0 vector with held-out $R^2 = 1.000$, against 0.126 for Pythia. GPT-2 adds learned absolute positional embeddings at the embedding layer, while Pythia's rotary encoding is applied inside attention, so mean pooling leaves a deterministic length term in one and not the other. Predicting the model that has it, from the model that does not, fails - correctly.

9. Reading a heat map honestly

A heat map invites a verdict by eye, and the eye is generous - it finds diagonals in noise, especially when one is expected. The CKA panel in Experiment A was described as showing a hot diagonal. Measured, the structure turned out to be different in three ways, one of which was stronger than the original claim.

1. Some evidence needs no correction - find it first. The diagonal entries rise strictly monotonically across nine consecutive layers (0.125 to 0.742, Spearman $\rho = 1.000$, $p < 10^{-5}$). No normalization, no modelling, no argument - and it matches the stitching curve from a methodologically independent stage. When a dataset contains a claim that survives with zero preprocessing, that claim is the one to lead with.

2. Distinguish the pattern from the alternatives, and calibrate on known answers. Four statistics separate a genuine diagonal from a banded one:

Statistic	True diagonal	Block pattern	Observed
mean argmax - layer index	0.00	6.00	2.85
diagonal / off-diagonal ratio	2.55	1.12	1.20
corr(similarity, proximity)	+0.83	-0.08	+0.131
permutation p on the diagonal	0.000	0.091	<0.001

The first two columns are not guesses - they are the same statistics computed on synthetic matrices constructed to contain each structure, so the diagnostic was calibrated before being trusted. The observed matrix sits between them.

3. Ask what else could produce the appearance. The blocky look was a marginal effect, not a pairing. Row means spanned 0.191 to 0.572 and a single rank-one component explained 97 per cent of the matrix energy - meaning most cells were predicted by 'how similar is this layer to everything' rather than by which layer it was paired with. Dividing out the row and column marginals (the standard lift correction) raised the proximity correlation from +0.131 to +0.242 and the share of layers matching within two from 62 to 77 per cent. The correspondence was there; a nuisance structure was sitting on top of it.

4. Expect measures to disagree, and locate the property that explains both. CKA is invariant to rotation, permutation and isotropic scaling but not to anisotropy, which ridge regression absorbs. Given the effective dimensionalities in Section 8, a hubness-dominated CKA matrix beside a stitching R^2 of 0.737 is the predicted signature of matched content in mismatched variance spread. Disagreement between two measures is information, not noise.

The general rule. A fitted map is never trusted on in-sample scores, and a figure should not be trusted on visual impression. Hold claims about pictures to the standard already applied to claims about numbers: state the expected pattern, compute statistics that would distinguish it from the alternatives, calibrate them on cases with known answers, check what nuisance structure could imitate the pattern, and report what survives.